

UST-OLENEK, Russian Federation

WMO: 217110

Lat: 73.000N

Lon: 119.867E

Elev: 49

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-46.6	-44.5	-51.3	0.3	-46.0	-49.1	0.3	-44.2	27.7	-23.0	24.0	-25.3	9.7	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	69.0	56.8	64.2	54.2	59.4	51.3	58.1	66.7	54.6	62.9	52.1	59.0	9.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.8	61.7	63.2	49.8	53.0	58.0	47.0	47.8	56.0	25.2	67.4	23.0	63.6	21.5	58.9	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.0	19.9	DB	-48.5	73.4	4.1	6.3	-51.5	77.9	-53.8	81.5	-56.1	85.0	-59.1	89.6	(4)
(5)			WB	-48.3	60.1	3.7	2.4	-51.0	61.8	-53.1	63.2	-55.2	64.5	-57.8	66.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-25.0	-26.3	-10.6	5.1	21.5	40.8	46.2	47.8	36.8	15.5	-8.9	-20.8		
	DBStd	29.09	11.95	11.94	14.82	11.01	9.09	8.76	8.16	6.26	6.32	10.40	12.19	11.21		
	HDD50	14602	2326	2136	1880	1346	884	303	175	121	399	1068	1768	2196		
	HDD65	19945	2791	2556	2345	1796	1349	728	586	535	847	1533	2218	2661		
	CDD50	138	0	0	0	0	0	27	58	51	2	0	0	0		
	CDD65	5	0	0	0	0	0	1	4	0	0	0	0	0		
	CDH74	65	0	0	0	0	0	20	45	0	0	0	0	0		
	CDH80	13	0	0	0	0	0	5	8	0	0	0	0	0		
(14) Wind	WSAvg	9.8	10.3	10.1	10.0	9.7	10.3	10.3	9.8	9.5	9.2	8.6	10.0	10.6		
(15) Precipitation	PrecAvg	12.1	0.8	0.5	0.7	0.6	0.9	0.8	1.3	1.8	1.6	1.5	0.9	0.9		
	PrecMax	15.5	1.6	1.0	1.6	1.9	2.1	1.6	2.7	4.2	3.4	2.6	2.8	1.8		
	PrecMin	7.7	0.2	0.0	0.2	0.0	0.2	0.2	0.4	0.5	0.5	0.5	0.2	0.3		
	PrecStd	2.2	0.4	0.3	0.3	0.4	0.4	0.4	0.5	0.9	0.8	0.5	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	5.3	27.7	34.9	47.1	75.6	80.2	70.2	59.7	34.3	27.0	17.8		
		MCWB	14.5	4.0	26.3	31.9	40.1	58.0	65.9	57.6	51.6	32.5	25.6	17.2		
	2%	DB	1.7	0.3	23.3	30.2	41.9	66.7	70.9	66.2	51.2	32.8	21.8	3.2		
		MCWB	0.6	-0.7	21.8	27.7	36.2	53.8	60.4	54.8	46.0	31.0	20.5	2.4		
	5%	DB	-2.2	-4.8	17.9	26.4	37.3	60.1	65.1	61.7	47.7	30.8	13.8	-0.6		
		MCWB	-2.8	-5.6	16.6	24.3	34.5	50.2	55.4	53.5	43.9	29.1	12.7	-1.3		
	10%	DB	-8.6	-8.8	11.3	21.9	34.2	55.5	59.5	57.8	44.8	28.2	8.1	-4.6		
		MCWB	-9.1	-9.3	10.1	20.1	32.1	47.8	51.1	51.1	41.6	26.3	7.1	-5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	4.4	26.5	32.1	41.3	58.9	66.1	58.8	52.4	33.0	25.9	17.1		
		MCDB	15.6	5.2	27.7	34.3	46.6	74.7	79.9	66.8	58.3	33.9	27.1	17.8		
	2%	WB	0.9	-0.6	21.8	28.1	36.8	54.4	61.5	55.9	47.0	31.2	20.5	2.6		
		MCDB	1.6	0.0	23.2	30.0	41.1	65.1	70.2	63.3	50.2	32.6	21.8	3.1		
	5%	WB	-2.7	-5.4	16.5	24.3	34.3	51.3	55.8	53.9	44.1	29.1	12.7	-1.4		
		MCDB	-2.0	-4.9	17.9	26.5	37.0	59.6	63.0	61.2	47.2	30.7	13.6	-0.8		
	10%	WB	-9.1	-9.3	10.1	20.2	32.3	47.6	52.2	51.7	41.7	26.5	7.1	-5.2		
		MCDB	-8.7	-8.7	11.1	21.8	34.4	55.2	59.1	57.7	44.5	28.0	8.1	-4.5		
(35) Mean Daily Temperature Range	MDBR		9.5	9.3	11.7	14.5	10.1	12.5	11.8	10.7	6.5	8.2	10.1	10.0		
	5% DB	MCDBR	15.6	14.3	13.9	16.5	12.0	23.1	21.4	17.2	12.2	7.9	16.4	15.0		
		MCWBR	15.4	13.9	13.3	15.3	8.8	14.2	13.1	10.6	8.9	7.8	15.9	14.8		
	5% WB	MCDBR	15.2	14.1	13.8	15.8	11.3	22.9	20.2	15.8	10.7	7.9	16.3	15.1		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.185	0.274	0.310	0.341	0.339	0.376	0.370	0.301	0.198	N/A	N/A		
	taud		N/A	1.908	2.013	2.026	2.046	2.294	2.300	2.320	2.383	1.950	N/A	N/A		
	Ebn at Noon		0	123	206	246	255	264	246	229	210	106	0	0		
	Edh at Noon		0	9	24	35	40	33	32	27	17	8	0	0		
(44) All-Sky Solar Radiation	RadAvg		0	47	483	1285	1831	1840	1552	906	407	93	0	0		
	RadStd		0	5	26	49	113	137	107	111	45	14	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page